Maaz Qureshi

Research Engineer (Machine Learning, Physical AI: Robotics) : MASc Mechatronics, University of Waterloo

Portfolio | LinkedIn | Git | Google Scholar | +1 437 870 8791 | m23qures@uwaterloo.ca | Waterloo, Canada

Skills

Languages & Frameworks: ROS2 , Python , C++ , PyTorch, Eigen / Numpy , TensorRT (NVIDIA CUDA)

Machine Learning: VLA , CNN , VLM , Diffusion Policy , Quantization , Training (backpropagation) , Reinforcement Learning(optimization based)

Robot Devolvement: VI SLAM (Volumetric Mapping & Localization) , Motion Path Planning 7-DoF (Kinematics IK/FK) , Sensor Fusion (3D Stereo RGB-D Camera, LiDAR, mmWave 4D RADAR, IMU) , 6D Pose Estimation , 5G Connected Factory Robotics

Tools: Docker , CI/CD , gRPC , Nvidia Isaac , MuJoCo , Rviz , Gazebo , Software QA (Debugging: Valgrind, GDB; Monitoring: Jtop)

Experience

Embodied AI Researcher: May 2026 – Present

Apple *Waterloo, Canada*

- Confidential-NDA project. Developing the research on VLAs (7 Dof Manipulator) for Apple Robotics with University of Waterloo.
- Developed the iPhone Pro ARkit for human Ego-Centric Data to fine tune the foundational models: Vision Language Action (pi family).
- Benchmarking the Ego-Centric Data with Franka Manipulator 6D pose + gripper width sensing as ground truth with MSE less than 10%.

Robotics Software Engineer: video Sept. 2025 – April 2026

Cobionix Corporation Inc *Kitchener, Canada*

- Developed 22 FPS real-time RGB-D Multimodals AI pipeline (SMol_VLM, pose estimation, object detection) with C++ Nvidia CUDA.
- Quantization with TensorRT (FP16, INT8) for async pipelines to maximize throughput (increase camera FPS, reduce latency) to run parallel.
- Developed live RGB-D stereo sensor fusion algorithm with intrinsic/extrinsic (temporal and spatial 90%+) calibration pipeline.

Graduate Research Student: ML and Robotics Manipulation-Perception Jan. 2024 – Aug. 2025

University of Waterloo RoboHub Lab [Publication, Video, GitHub] *Waterloo, Canada*

- Developed novel distributed multi-robot dense volumetric SLAM (C++/ROS2 humble) with TSDF-octree + ORBSLAM3, achieving 85% improvement over state-of-the-art baselines for high-fidelity 3D scene reconstruction with 4 AMRs (mobile robots).
- Integrated perception pipeline for cluttered settings using mmWave 4D RADAR + RGB-D, achieving 92% accuracy in detection.

Keysight Technologies Inc [Publication, Video, GitHub] | MITACS Research Internship

- Developed motion planning pipeline (OMPL + RRT) for 7-DoF Franka/KUKA collaborative robot, SIM-to-Real deployment.
- Achieved 100% joint-limit and collision avoidance across hemispherical pattern formations. Used python with ROS rviz, and docker setup.

Rogers Communications Inc | MITACS Research Internship

- Integrated 5G in DISTRIBUTED software pipeline on NVIDIA Jetson for smart factory robots achieved 37% latency reduction.
- Benchmarked 5G sub-6 GHz (3.5 GHz) and mmWave (28 GHz) channels against WiFi for industrial robotics to increase bandwidth.

Robotics Asset Integrity Engineer: May 2022 – Dec. 2023

ABYSS Solutions Inc *Islamabad, PK (Onsite) & Remote Australia*

- Performed Visual-SLAM (mapping) perception models for oil-gas mooring chain + DAMs inspection with Under Water Robot (ROV) data.
- Contributed to robotics software development for ROS to ROS2 AMR pipelines, VR for ADIPEC, and agriculture quadcopter projects. [Link]
- Led a project, 10-engineer team on digital inspection, 18,000+ assets with 91% coverage for a BP offshore rig and Dam Data-Capture.[Link]

Side Projects

- **Humanoid Robotics – Non-Verbal Human-Robot Interaction** [Link]: NAO robot in kitchen with humans for breakfast preparations.
- **Visual Features with Machine Learning:** KNN, SVM, and FRIQUE image quality benchmarking for computer-generated vs real-world data.
- **AMRs in Health Care** [Link]: RFID nursing robot for pill dispensing, perception, and live stream in healthcare environments.

Education

University of Waterloo | Thesis Research: AMRs 3D SLAM Link1 , 7 DoF Manipulator Motion Planning Link2 Waterloo, Canada

Master of Applied Science (MASc), Mechatronics Engineering ***(CGPA: 93.5%)*** January 2024 – August 2025

Supervisor(s): Dr. William Melek ; Dr. George Shaker

Bahria University | Thesis: AMRs Swarm Robotics Autonomy and Perception Islamabad, PK

Bachelor of Science (BSc), Electrical Engineering ***(CGPA: 80.5%)*** September 2017 – August 2021

Selected Publications

- **M. Qureshi et al.**, "CRADMap: Applied Distributed Volumetric Mapping with 5G-Connected Multi-Robots and 4D Radar Perception," *IEEE ICARM 2025*. [Link]
- **M. Qureshi et al.**, "Automated Angular Received-Power Characterization of Embedded mmWave Transmitters Using Geometry-Calibrated Spatial Sampling," *IEEE Transactions on Microwave Theory and Techniques (T-MTT)*, 2026. [arXiv]

Achievements

- **Employee Recognition:** Best Performer Employee of the Month, Abyss Solutions Inc (industry).
- **Innovation Awards:** Winner of 3 hackathons in robotics and VR innovation at Abyss Solutions.
- **Academic Excellence:** Awarded full MASc graduate funding and recognized among the top 3 undergraduate thesis projects.